# Ayush Gupta

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## EDUCATION

| Course | College / University | Year | CGPA / % |
|---|---|---|---|
| B. Tech (Electrical Engineering) | Netaji Subhas University of Technology | 2027 | 7.73 |
| CBSE (Class XII) | Navyug Senior Secondary School | 2022 | 86% |
| CBSE (Class X) | DAV Public School | 2020 | 95% |

## EXPERIENCE

**Software Engineer Intern** | Bachatt *May 2026 – July 2026*
*Certificate*

- Stood up production infrastructure on Azure AKS (Helm charts, cluster bootstrap, secrets) and closed a 2x app load-time regression that followed a CDN migration, compressing 134 live assets (117 images plus 17 GIF-to-WebP conversions) for 47 MB freed at 75% average reduction, republished to both AWS S3 and Azure Blob.
- Built the daily credit-card MIS funnel report that runs inside the team's internal automation platform: pulls each partner's MIS from the mailbox, normalises ZET, Uni and Bajaj into one combined workbook and emails it on a schedule, so the funnel numbers arrive without anyone assembling them by hand.
- Turned a one-off asset cleanup into a standing guarantee: replaced an opt-in compress checkbox with an enforced 100 KB image gate that auto-compresses in a blocking dialog, closed the loophole where zipping oversized images bypassed it by scanning archive entries before extraction, and rejected video uploads whose adaptive bitrate ladder was missing rungs.
- Kept shipping into the self-hosted forks after the migration rather than living with upstream defaults: single-assignee enforcement in the tracker backend, separate cycle start and end dates, a join-a-project onboarding step for invited users, and inline PDF rendering with a selectable text layer in the wiki.

## PROJECTS

**Findesk — AI Equity Research & Portfolio Platform** | *Next.js 14, TypeScript, Python, Gemini API, Supabase* *Live*

- Shipped 0-to-1 to six live surfaces: stock research (price, fundamentals, news, AI Buy/Sell brief with a 12-month target), a mechanical NIFTY-500 value+momentum screen, a portfolio tracker with an AI health analyzer, a paper-trading desk, compare and watchlist, on Supabase auth with row-level security; verified behavior with 752 automated tests.
- Built a portfolio-aware chat assistant on Gemini function calling with 4 server-side tools so the model fetches only what a question needs; user identity comes from the session cookie and never the request body, so a crafted request cannot read another user's holdings.
- Added deterministic evidence guards for unknown analyst coverage, headline-only news causation, stale quotes and unsupported feed provenance; withheld model preambles during tool calls and verified the changes with regression tests and desktop/mobile rendering checks.

**Project Aether — Offline Disaster Mesh** | *Python, FastAPI, libp2p, React* *GitHub*

- Chose a grow-only set CRDT as the ledger so merges are commutative, associative and idempotent - two nodes reconcile correctly whenever they happen to meet, in any order, after any gap; identity is (node_id, seq) and ordering is a Lamport clock, because offline devices have no synchronised clock and wall-clock time can never be trusted as merge order.
- Implemented libp2p transport with mDNS discovery, gossipsub broadcast and anti-entropy catch-up. In single-machine multi-node tests, three nodes converged in about 2 seconds from cold start; partitioned nodes diverged, then reconverged in about 1 second after healing.
- Ran a chaos matrix against the real code rather than mocks - clock skew, duplicate injection, concurrent ingest, SIGKILL mid-write and a hung peer - which found and fixed two genuine defects a normal test suite missed: a thread-safety race on ingest and a missing read timeout on the sync path.

**Scrum Dashboard — Sprint Health on Plane** | *Next.js, TypeScript, Prisma/Postgres* *Live*

- Split identity from authorisation because they could not be the same thing: Google SSO restricted to the company domain answers who you are, while each user links their own tracker access token - validated before storage and encrypted at rest - so the dashboard shows each manager exactly what their own permissions allow and nothing more.
- Designed the snapshot job around a constraint that cannot be undone: a missed day is a permanent hole in the burndown, so charts render gaps as gaps rather than interpolating over them, a repeat run is a no-op rather than a duplicate row, and the job reports which cycles failed instead of failing silently.

## ACHIEVEMENTS & CERTIFICATIONS

- Published a peer-reviewed research paper on solar PV performance enhancement in Physica Scripta (IOP Publishing).
- Solved 600+ data-structures and algorithms problems on LeetCode; Pupil on Codeforces.
- Gold Medallist, NPTEL Programming in Java (IIT-certified).

## SKILLS

**Engineering:** TypeScript/JavaScript, Python, C++, Next.js, React, Node.js, FastAPI, Prisma/Postgres, REST APIs

**AI & Systems:** LLM function/tool calling, fixed-snapshot AI evaluation & model benchmarking, CRDTs (grow-only sets, merge semantics), libp2p (mDNS, gossipsub, anti-entropy), partition tolerance & chaos testing

**Infrastructure:** Docker, Kubernetes (Azure AKS), Helm, Jenkins CI/CD, Git, Supabase